Yazgi Akata

Doctoral Student

Graduate Research Assistant at Virginia Tech Institute for Advanced Computing — Arlington, VA
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ABOUT

I am broadly interested in artificial intelligence for social impact, especially how computational methods can address social challenges. I am particularly interested in exploring the trade-off between personalization and content diversification in recommendation algorithms, drawing on methodologies from computer science and the social sciences, and developing algorithmic interventions to enhance constructive cross-community interactions.

EDUCATION

Virginia Tech, Arlington, VA — August, 2026 — Present
Doctor of Philosophy in Computer Science advised by Dr. Sanmay Das

The State University of New York at Buffalo, Buffalo, NY — August, 2022 — May, 2026
Bachelor of Science in Computer Science — Minor in Philosophy, Politics and Economics — Cumulative GPA: 3.7/4.00
Yildizlar Education Fund Scholar

Darussafaka Schools, Istanbul, Turkiye — September, 2014 — June, 2022
Valedictorian

SCHOLARSHIPS & AWARDS

Faculty Choice University at Buffalo Department of Computer Science & Engineering (May 2026) Buffalo, NY
Awarded to one or more graduating seniors who have been a most vital presence in the department and whose departure will be sorely missed, not only by the faculty, at whose discretion the prize is awarded, but by the department as a whole.

Yildizlar Education Fund Scholar (2022-2026) Buffalo, NY
Scholarship to SUNY at Buffalo covering full tuition, board, flights and stipend provided to gifted, but underprivileged students.

Darussafaka Schools Student Scholarship (2014-2022) Istanbul, Turkiye
Full scholarship to Turkiye's first educational NGO, which provides quality education and boarding opportunity to talented but financially disadvantaged kids with deceased fathers and/or mothers. Darussafaka had 0.06% acceptance rate in 2014.

Kennedy-Lugar Youth Exchange and Study Program (2019-2020) NJ, USA
Funded by the United States Department of State, scholarship offered to students from countries of strategic importance to live and study in the US for an academic year to promote mutual understanding among cultures.

FIRST Robotics Competition World Championship Engineering Inspiration Award (2019) Houston, TX
First team from Turkiye to be globally recognized at the world championship for demonstrating outstanding success in advancing respect and appreciation for engineering within a team's school or organization and community.

PROJECTS

Kademlia Buffalo, NY

- Inspired by Kademlia, implemented a peer-to-peer distributed hash table (DHT) in Go, utilizing XOR based node lookup for routing table maintenance.

CSE Interview Help Tool Buffalo, NY

- Cleaned, processed and analyzed approximately 25,000 lines of student interview data using data analysis tools and TF-IDF method in Python. Trained artificial intelligence driven tool using KNN classification to help students improve their technical communication skills and success in assessments.

Spinderfy Buffalo, NY

- Designed and developed both the front and back end of a web application that allows the users to like and skip music, recommended based on user's past behavior, using Figma, React, Vite and PHP. Integrated Spotify and Gemini API into the process.

EXPERIENCE

Virginia Tech Institute for Advanced Computing Arlington, VA

Graduate Research Assistant advised by Dr. Sanmay Das August 2026 — Present

- Conducting exploratory research in computational social science and AI for social impact, analyzing sensitive administrative data using Python (NumPy, pandas, scikit-learn).

Department of Computer Science & Engineering, SUNY at Buffalo Buffalo, NY

Undergraduate Student Researcher advised by Dr. Atri Rudra January 2025 — May 2026

- Conducted research under Dr. Atri Rudra on the reliability of a linear algebraic operator simulating gated-convolution architectures. Proposed a parity-based error-correction method that adds auxiliary data into the input and model parameters, and recovers corrupted outputs without recomputation, or restructuring the operator. I am currently formalizing the approach.

Head Undergraduate Teaching Assistant May 2024 — January 2026
CSE116: Introduction to Computer Science II

- Conducted the leveraging of a tool used for streamlining the process of an assessment used to assess technical communication skills of students. Assembled questions targeting students' understanding, and oversaw the grading of these assessments effectively communicating with both instructors and teaching assistants. Completed administrative tasks.

Undergraduate Teaching Assistant
CSE486: Distributed Systems January 2026 — May 2026
CSE220: Systems Programming August 2024 — May 2026
CSE116: Introduction to Computer Science II January 2024 — January 2026
CSE115: Introduction to Computer Science I August 2023 — December 2023

- Addressed concepts such as compiler and linker invocation, memory management, threads, synchronization primitives, data structures, classes, and graph traversal algorithms by leading several weekly lab sections and office hours covering weekly lab assignments. Collaborated with 30+ course staff to ensure smooth operation of the course, contributed to the improvement of each course by consistently providing feedback during the weekly TA meetings, and proctored and graded assessments while ensuring academic integrity and fair opportunity.

SKILLS

- **Programming:** Python, C, Go, Java, JavaScript, PHP
- **Software:** Emacs, Visual Studio Code, IntelliJ Idea
- **Frameworks:** React, Django

OTHER EXPERIENCES & VOLUNTEERISM

UBHacking Buffalo,NY
Advisor March 2025 — May 2026
Director March 2024 — March 2025
Sponsorship Lead August 2023 — March 2024

- Directed UB Hacking, a 500+ participant weekend-long hackathon focused on computing for social good, coordinated workshops and judging with faculty, trained and managed an 80% new organizing team, oversaw fundraising and budgeting to ensure sustainability, advised subsequent organizers, and led sponsor communications and onboarding.

Campus Living at SUNY at Buffalo Buffalo, NY
Community Assistant at Hadley & South Lake Villages August 2024 — August 2025

- Fostered an inclusive and supportive living environment for 69 residents by organizing community-building events, providing day-to-day guidance, and addressing resident concerns with empathy and care.

International Student Advisory Council at SUNY at Buffalo Buffalo,NY
Founding Board Member October 2022 — May 2026

- Advocated for the international student body at UB, served as an accessible resource to students, provided guidance, assistance and feedback to the International Students Services office on student inclusion.

Young Guru Academy Hatay, Turkiye
Volunteer for World Science Movement January 2025

- Led workshops to encourage a love for science for underprivileged elementary school students in the still recovering areas from the earthquake of February 6th, 2023 in Hatay, Turkiye.

AFS Intercultural Programs Washington D.C.
Workshop Facilitator & Volunteer 2023—2025

- Facilitated workshops on cultural transition and intercultural communication for the incoming class of youth ambassadors from various countries for the KL-YES Program funded by the Department of State.

REFERENCES

Dr. Sanmay Das
Associate Director of AI for Social Impact at Virginia Tech, Arlington, VA
E-mail: sanmay@vt.edu
Dr. Atri Rudra
Katherine Johnson Chair in Artificial Intelligence, Department of AI and Society at SUNY at Buffalo, Buffalo, NY
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Dr. Ethan Blanton
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Dr. Paul Dickson
Assistant Professor of Teaching, Department of Computer Science and Engineering at SUNY at Buffalo, Buffalo, NY
E-mail: pauldick@buffalo.edu
Alan Hunt
Professor of Practice, Department of Computer Science and Engineering at SUNY at Buffalo, Buffalo, NY
E-mail: ahunt@buffalo.edu